A Floer-Theoretic Framework for Hamiltonian Bifurcations

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Based on a joint work with Hong-Kwon Jo (SNU), arXiv:2607.22178.

Main result. We describe the explicit changes of the local Floer chain complex for generic types of Hamiltonian bifurcations.

- **Motivation and Ideas.**
  Bifurcation and Floer homology.
- **Main Results.**
  Explicit changes of the local Floer chain complex across bifurcations.
- **Research Plans.**
  Toward more bifurcations, Floer-type Cerf theory.

Bifurcation and Floer Homology

Motivation. Bifurcations and Conley–Zehnder indices

A **bifurcation** is, heuristically, the birth or death of periodic orbits.

Several recent works [AFvK$^+$23], [AB25], [JKvK26] relate bifurcations to symplectic invariants, for example Conley–Zehnder indices.

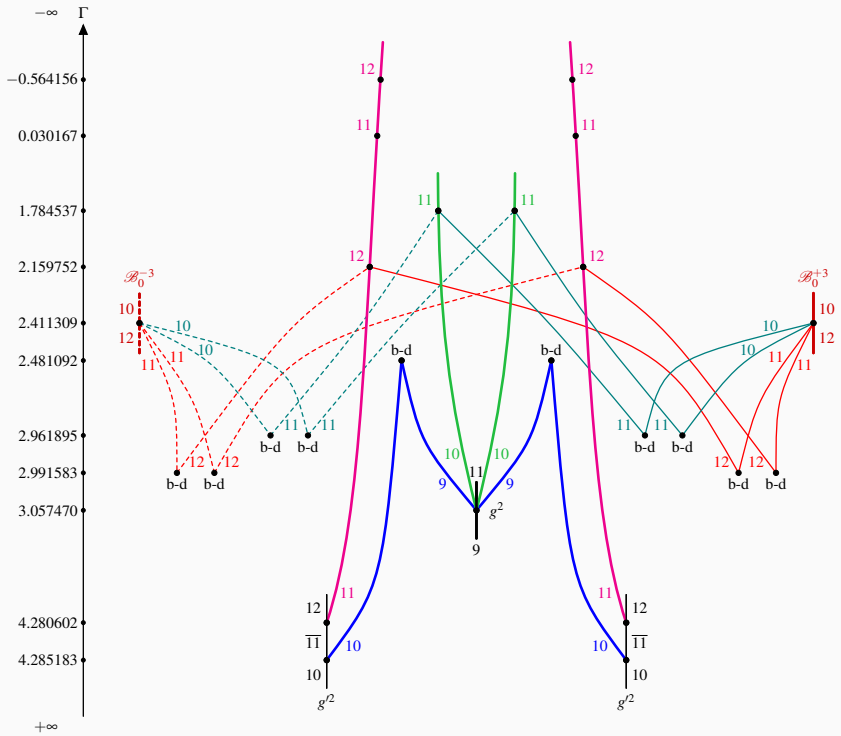


Figure 1: Bifurcation graph of Hill's lunar problem, [AB25].

Bifurcation

Setup. 4-dimensional autonomous Hamiltonian system with 1-dimensional bifurcation parameter ε, say $(W, \omega, H_\varepsilon)$.

Let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0.

- If $\lambda_0 \neq 1$, so γ_0 is non-degenerate, then γ_0 can be continued to small ε, so we get an **orbit cylinder**

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

  Then we follow γ_ε and detect λ_ε.
- When $\lambda_\varepsilon = 1$, the orbit cylinder breaks; new orbit might appear, or the family γ_ε might disappear. This is called **bifurcation**.
- If $\lambda_\varepsilon^k = 1$, γ_ε doesn't bifurcate but γ_ε^k does bifurcate.

Note. Conley–Zehnder index changes only when the orbit degenerates, and this fact is useful in practical purposes. ([AB25], [JKvK26].)

Example. Period Doubling

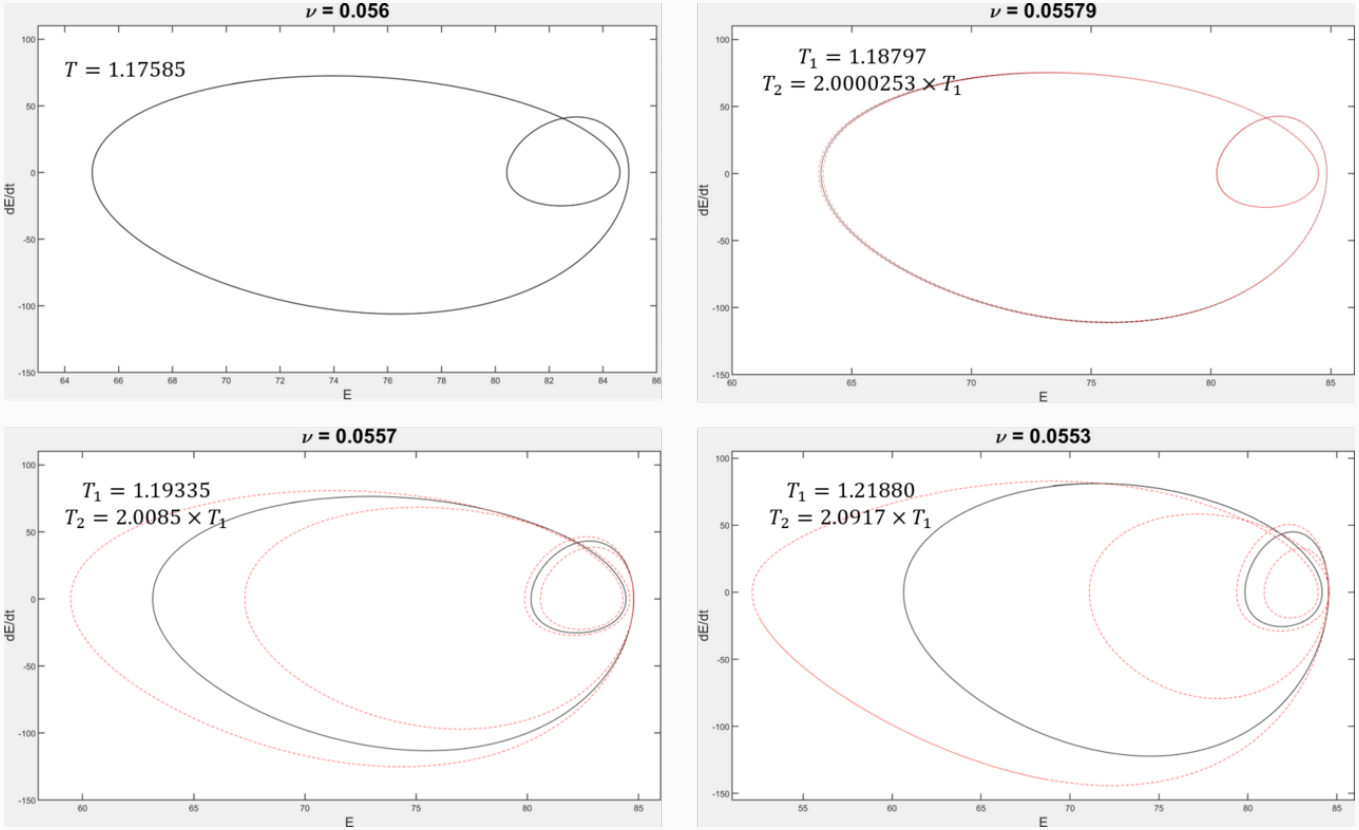


Figure 2: Period doubling in Kuramoto–Sivashinsky equation [Max21].

The red dashed orbit is the new orbit that emerges from the bifurcation.

This is a generic bifurcation of a double cover.

Floer Theory

The local Floer homology is invariant under small perturbation.

$\Rightarrow HF_*^{\mathrm{loc}}(W, H_\varepsilon)$ is invariant across bifurcation.

However, its chain complex CF_* changes substantially;

- orbits are born or vanish $\Rightarrow$ the number of generators changes.
- the CZ index changes $\Rightarrow$ grading changes.

In short, $CF_*(H_{-\varepsilon})$ and $CF_*(H_\varepsilon)$ are different chain complexes, but their homology $HF_*(H_{-\varepsilon})$ and $HF_*(H_\varepsilon)$ are the same.

$\Rightarrow$ The Floer differential is responsible for keeping the homology invariant. We call this **mutation** of LFCC.

Main Goal. Describe this effect explicitly for generic bifurcation types.

Main Results

Outline

We're interested in the case such that $\lambda^k = 1$.

1. By **Birkhoff normal form**, find a generating Hamiltonian G_ε of Φ^k.
2. Construct **Model Hamiltonian** H_ε from G_ε.
3. Using **Floer cascade** to compute Floer differential.

Birkhoff Normal Form

- (W, H_ε): 1-parameter family of 4-dim Hamiltonian systems.
- γ_0: periodic orbit of H_0 with $\lambda_0^k = 1$.
- Φ_ε: Poincaré return map of γ_0, extended to $\varepsilon \neq 0$.

Key Idea. There exists an *autonomous* Hamiltonian G_ε consists of $\mathbb{Z}_k$-symmetric polynomials, which generates Φ_ε^k, the k-th iterate of Φ_ε.

$$G_\varepsilon = \begin{cases} \alpha u^2 + \beta v^3 + \varepsilon(au + bv + \ldots) & k = 1, \\ \alpha u^2 + \beta v^4 + \varepsilon(av^2 + \ldots) & k = 2, \\ \alpha \rho^4 + \beta \rho^k \cos k\varphi + \varepsilon(a\rho^2 + \ldots) & k \geq 3. \end{cases}$$

Generic condition: Leading coefficients are nonzero.

Critical points of G_ε corresponds to the k-periodic orbit of H_ε.

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70], which means the non-zero leading coefficients.

They are organized by the covering number k of the underlying orbit.

- $k = 1$: birth-death
- $k = 2$: period doubling
- $k = 3, 4$: phantom kiss
- $k \geq 4$: emission

Note. The fourth cover is a borderline case, depending on the coefficients of the normal form.

Model Hamiltonian

Let $\tilde{H}_\varepsilon$ be a Hamiltonian and G_ε be a generating Hamiltonian of Φ_ε^k.

We construct a model Hamiltonian ona neighborhood of γ_0, which is $I_r \times (\mathbb{R}/k\mathbb{Z})_\theta \times \Sigma_z$, by

$$H_\varepsilon = kh(r) + G_\varepsilon(z).$$

where $h'(0) = 1$ and $h''(0) \neq 0$.

- k-to-1 correspondence between $\mathrm{Crit}(G_\varepsilon)$ and $\mathrm{Per}(H_\varepsilon)$, say $p \leftrightarrow \gamma_p$.
- $\mu_{CZ}(\gamma_p) = 1 - \mathrm{Ind}_{\mathrm{Morse}}(p)$.
- A gradient flow between p and q gives a Floer cylinder between γ_p and γ_q, which we call a **gradient revolution**.

Note. H_ε and $\tilde{H}_\varepsilon$ are different, but has the same return map, bifurcation type, and Floer-type data up to finite degree and symplectomorphism.

Floer Cascade

Our system is autonomous, so we need to consider Morse–Bott type cascades to compute the differential between $\hat{\gamma}$ (max) and $\check{\gamma}$ (min).

Floer cascade is a sequence of Morse trajectories on the orbit and Floer cylinders between orbits.

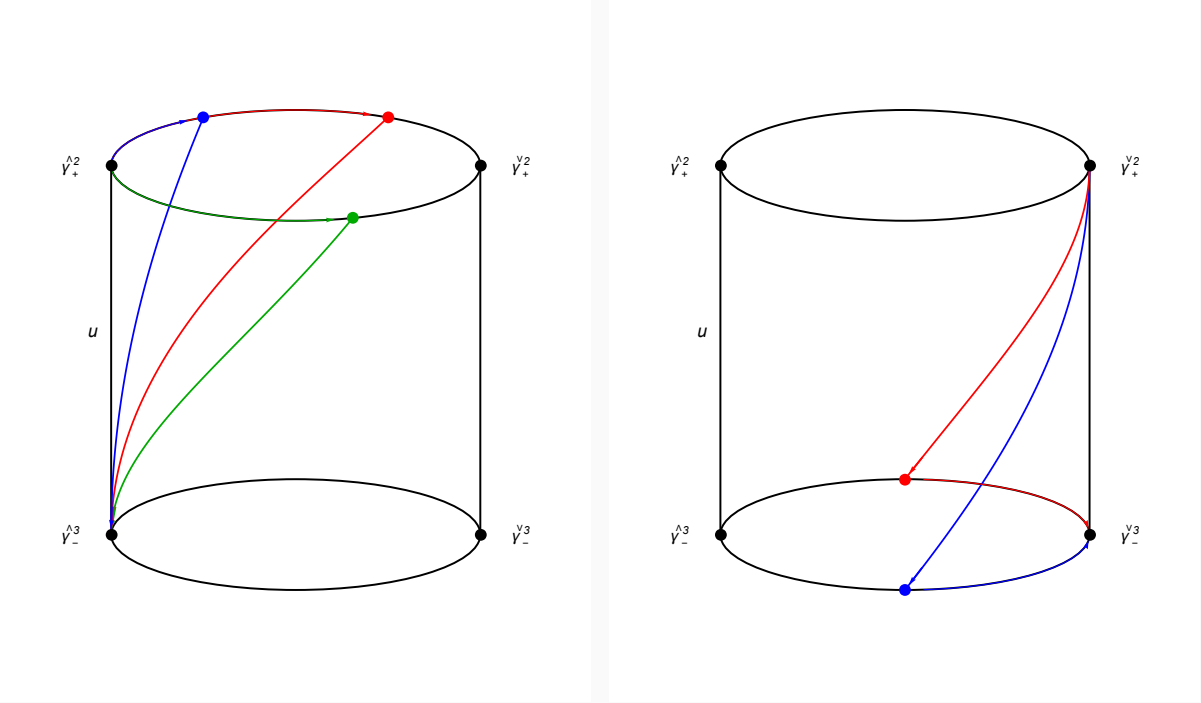


Figure 3: Cascades between 2nd and 3rd covers.

Cascade Counting Theorem

Theorem

Let $\gamma_\pm$ be two non-constant $1/k_\pm$-periodic orbits, so that $\gamma_\pm^{k_\pm}$ are 1-periodic orbits. Moreover, assume that

$$\mu_{CZ}(\gamma_+^{k_+}) - \mu_{CZ}(\gamma_-^{k_-}) = 1.$$

Then, the quotient $\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1$ is a 0-dimensional $\mathbb{Z}_{(k_+,k_-)}$-orbifold, and we have

$$\#_2\mathcal{M}(\hat{\gamma}_-^{k_-}, \hat{\gamma}_+^{k_+}) \equiv k_- \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod 2,$$
$$\#_2\mathcal{M}(\check{\gamma}_-^{k_-}, \check{\gamma}_+^{k_+}) \equiv k_+ \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod 2.$$

In particular, if there is a cylinder from a simple γ_0 to a multiple cover γ_1^k,

$$\partial\hat{\gamma}_0 = k\hat{\gamma}_1^k, \quad \partial\check{\gamma}_0 = \check{\gamma}_1^k.$$

Birth-death $(k = 1)$

A generic single cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^3 + \ldots + \varepsilon(u + \ldots).$$

- $(\varepsilon < 0)$ There are no periodic orbits.
- $(\varepsilon > 0)$ One elliptic γ_0 and one hyperbolic orbit γ_1 are born.
- There is one gradient trajectory between them.

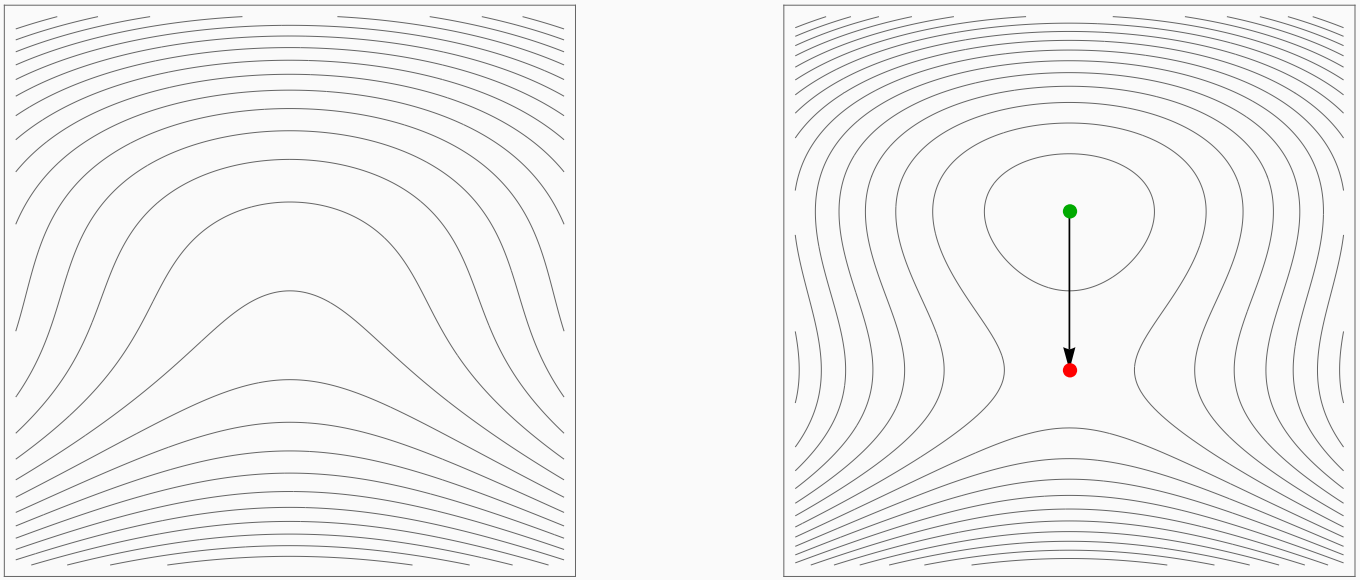


Figure 4: Critical points and gradient trajectories of birth-death bifurcation

Birth-death $(k = 1)$

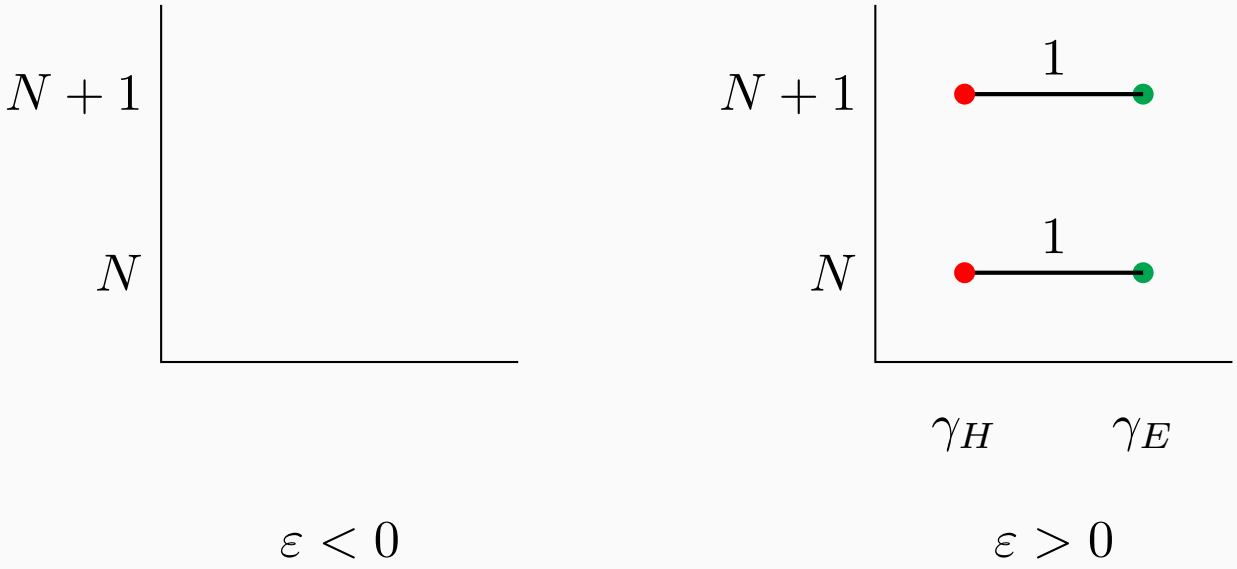


Figure 5: Chain complex mutation across the birth-death.

- Points at (n, m) corresponds to a generator of degree $n + m$.
- The line indicates the differential; $\partial(\hat{\gamma}_E) = \hat{\gamma}_H$ here.
- We are using $\mathbb{Z}_2$-coefficients.

We have $HF_*^{\mathrm{loc}}(\gamma) = 0$, where γ is the degenerate orbit at $\varepsilon = 0$.

Period doubling $(k = 2)$

A generic double cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^4 + \ldots + \varepsilon(u^2 + \ldots).$$

- $(\varepsilon < 0)$ One elliptic orbit γ_0^2 exists.
- $(\varepsilon > 0)$ γ_0^2 becomes hyperbolic, and an elliptic orbit $(\bullet)$ γ_{new} is born.
- There is one cylinder between γ_0^2 and γ_{new}.

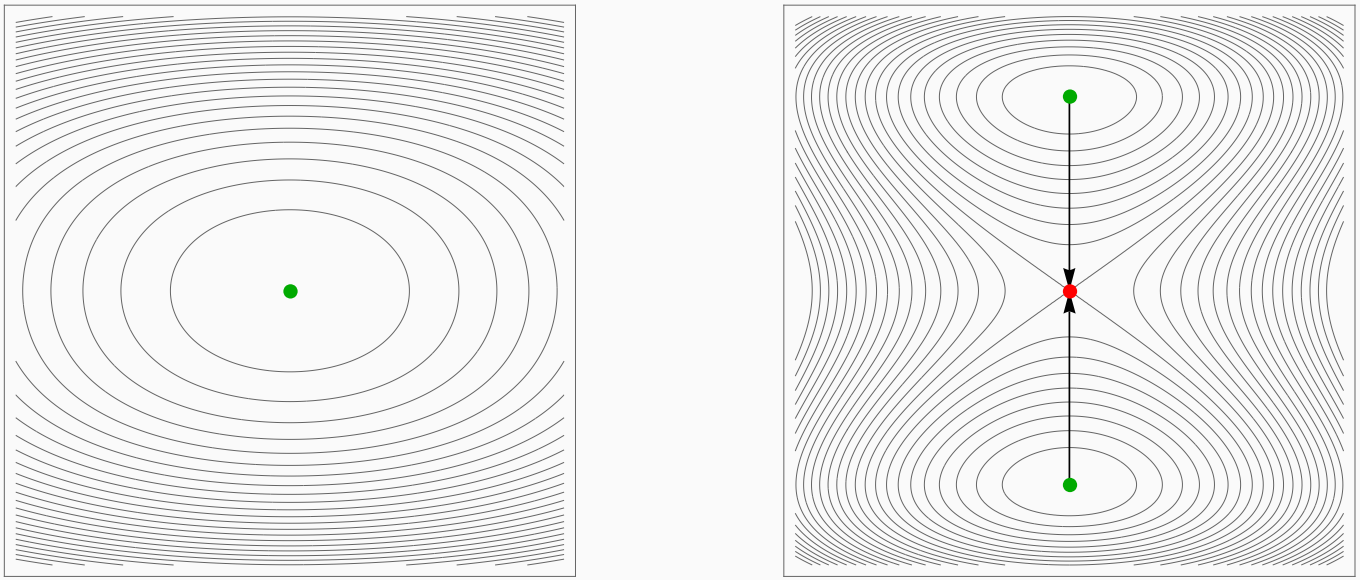


Figure 6: Period doubling bifurcation

Period doubling $(k = 2)$

- $\mu_{CZ}(\gamma_+^2; \gamma_-^2) = -1$ or 1. We assume -1 below.
- $\mu_{CZ}(\gamma_{new}; \gamma_-^2) = 0$.

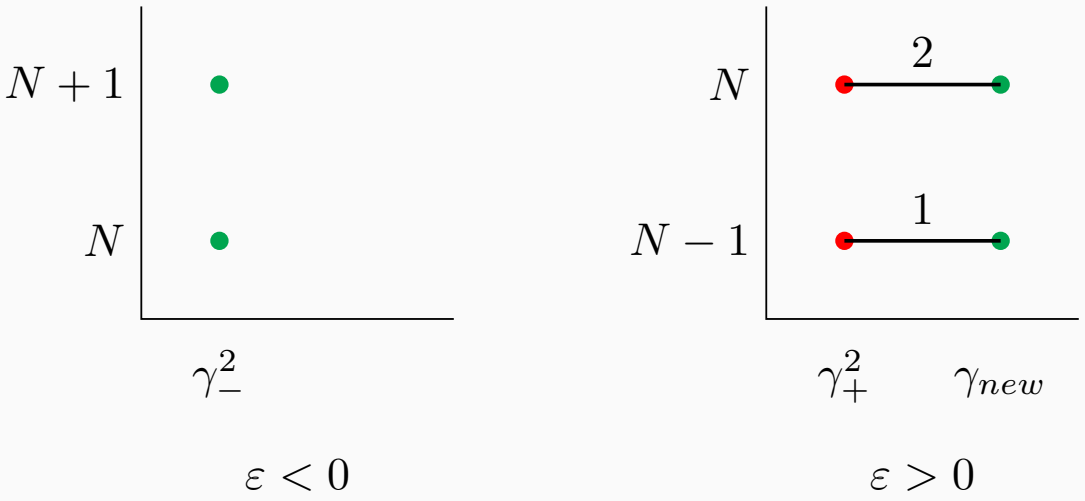


Figure 7: Chain complex mutation across the period doubling. $\check{\gamma}_H$ and $\check{\gamma}^2$ cancel each other, while $\hat{\gamma}_H$ and $\hat{\gamma}^2$ do not.

$$HF_*^{\mathrm{loc}}(\gamma_0^2) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^2, \\ 0 & \text{otherwise} \end{cases}$$

Research Plan

Symmetric Bifurcations and Lagrangian Floer theory

There are another types of bifurcations if the system has additional symmetries.

- CRTBP has one anti-symplectic involution and one symplectic involution, which makes the system D_2-symmetric.
- Hill's lunar problem has $\mathbb{Z}_2 \times D_2$-symmetry.
- Hénon–Heiles system has D_3-symmetry.

First example is the **pitchfork bifurcation**, which occurs for the simple orbit instead of the birth-death bifurcation.

Our previous work [JL26] covers some cases with $\mathbb{Z}_2$-symmetry, and we will extend this framework to broader classes of symmetric systems.

For anti-symplectic involution, Lagrangian Floer theory would also play a role: half-orbit is Lagrangian chord of the fixed locus.

Symmetry-breaking

Let there be an additional parameter μ such that $H_{\varepsilon,0}$ is G-symmetric but $H_{\varepsilon,\mu}$ is not for $\mu \neq 0$.

- At $\mu = 0$, symmetric bifurcation happens.
- At $\mu \neq 0$, this breaks into several generic bifurcations.

We expect that this would give some exact sequence between LFCCs, and provide an entrance to the multi-parameter bifurcations.

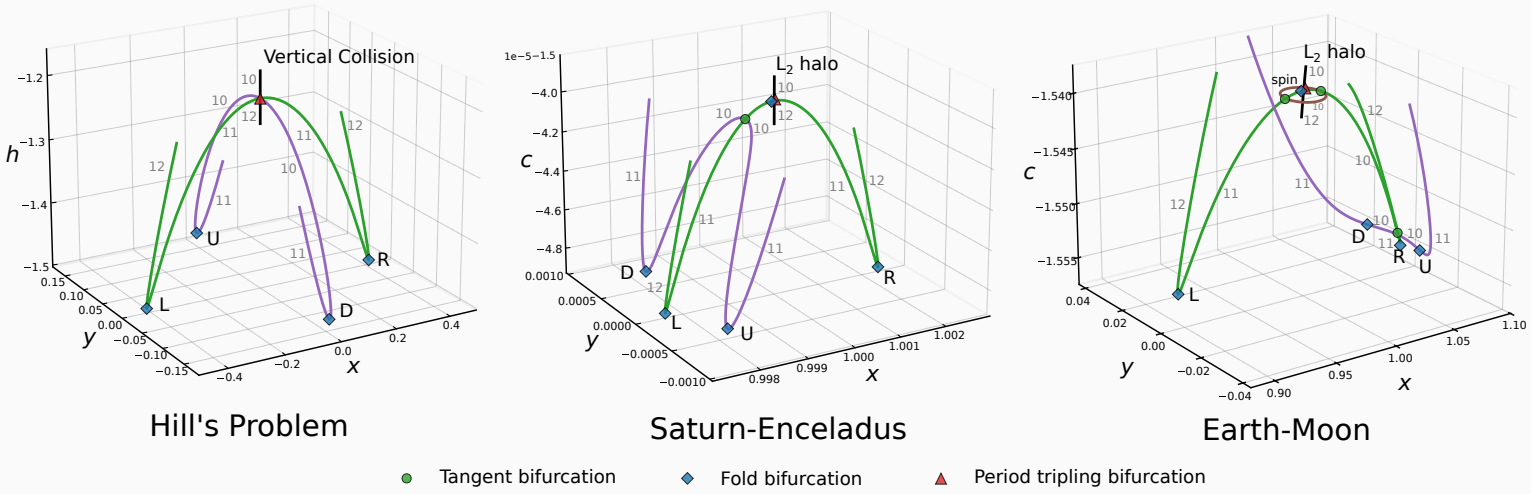


Figure 8: Symmetry-breaking, [JKvK26].

Persistence Modules and Floer–Cerf Theory

We can regard $CF_*(H_\varepsilon)$ as a 1-parameter family of chain complexes, suggesting a persistence-type structure.

Cerf theory describes how Morse complexes change along a parameter. What is the corresponding picture in Floer theory?

- We have the local ingredients: the classification of bifurcations and the corresponding changes of the LFCC.
- How do the moduli of non-local Floer cylinders change? Not clear.

Goal: assemble these local mutations into a global Floer-theoretic analogue of Cerf theory.

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Illustration of period doubling in the Kuramoto–Sivashinsky equation. CC0 1.0 Universal Public Domain Dedication.



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Figure 9: Thank you for your attention!

Questions, comments, and further discussion are welcome!